

Semantic Enrichment in Few-Shot Learning

Bridging the gap between numeric IDs and Foundation Models

Delucchi Tommaso, Mirafiori Elia, Segalla Filippo, Pettene Vittorio

June 10, 2026

University of Verona / Department of Engineering for Innovation Medicine

Table of Contents

- Introduction
- Scientific Inspiration
- Proposed Method
- Results and Challenge Outcome
- Conclusion and Future Work
- Individual Contribution

Introduction

- **What is Few-Shot Learning (FSL)?**
 - Classifying images from categories **completely unseen** during the main training phase.
- **Our Target Setup: 5-Way 5-Shot Classification**
 - The model must dynamically adapt to **5 novel classes**.
 - Exactly **5 labeled examples (Shots)** are available per class in the Support Set.
- **Why use FSL with Foundation Models?**
 - Eliminates massive data annotation costs.
 - We exploit the massive prior knowledge of internet-scale pre-trained backbones.

- **The Dataset Constraint**

- Classes do not have real names (e.g., "Kitchen"). They are provided purely as **anonymous numerical IDs** (e.g., Class 0, Class 1, Class 2...).

- **The Blind Spot for Vision-Language Encoders**

- Multimodal models like CLIP require natural language to align text and vision spaces.
- Prompting text encoders with "*a photo of a [1]*" is semantically empty.
- **Result:** Standard Zero-Shot CLIP performance collapses.

- **Nature of the Images**

- The data focuses on complex **indoor and outdoor environments** rather than isolated, centered objects.
- Categories include: *Corridor, Staircase, Patio, Living Room.*

- **Technical Implications**

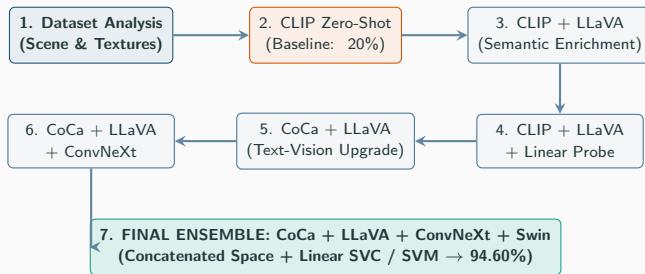
- Background textures and global layout constraints are vital.
- Dictates the adoption of advanced visual backbones (**Swin Transformer** & **ConvNeXt**) to capture rich spatial details.

Target Domain

Complex Scenes

Texture-heavy backgrounds & widespread layout details.

Introduction



Key Takeaway: Moving from a semantic-blind zero-shot baseline (2) to an incremental multi-backbone concatenation (6), finalized by spatial enrichment (Swin) and maximum margin regularized classification (7).

Scientific Inspiration

The "Embedding" Strategy

Paper: *Tian et al., "Rethinking Few-Shot Image Classification" (2020)*

Inspiration: Instead of using complex meta-learning algorithms, this paper proved that extracting a **high-quality representation** (embedding) is the most critical factor for success.

Key Phrase: "Good embeddings are all you need."

Solving the Semantic Gap

Paper: *Liu et al., "Visual Instruction Tuning (LLaVA)" (2023)*

Inspiration: We used LLaVA's ability to "reason" about images to translate our anonymous numeric labels into **meaningful semantic descriptions**.

Key Phrase: "Contextual reasoning for numeric tags."

Beyond Standard CLIP

Paper: *Yu et al., "CoCa: Contrastive Captioners are Foundation Models" (2022)*

Inspiration: CoCa is strictly stronger than CLIP because it doesn't just "match" images and text; it is forced to **describe them**, capturing much finer visual details.

Key Phrase: "Generative details + Semantic vision."

Uniting CNNs and Transformers

Paper: *Peng et al., "Conformer: Local Features Coupling Global Representations" (2021)*

Inspiration: We concatenated CNN and Swin features because this study proves that merging **local convolutional details** with **global transformer context** drastically maximizes accuracy.

Key Phrase: "Local CNN features + Global Transformer context."

Maximizing the Margin

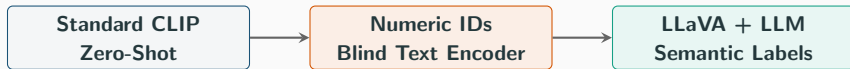
Paper: *Chen et al., "A Closer Look at Few-Shot Classification" (2019)*

Inspiration: This study shows that simple linear classifiers (like our **SVM**) often beat complex algorithms when the concatenated features are strong and high-dimensional.

Key Phrase: "Maximum margin over complex loops."

Proposed method

Proposed Method



The Setup & The Flaw

We initially deployed CLIP to classify images directly. However, because our dataset relies purely on anonymous numerical IDs (e.g., "Class 1"), CLIP's text encoder was completely blind. This broke the natural language connection, resulting in a severe performance drop.

The Solution

To restore the model's alignment, we introduced **LLaVA**. By visually analyzing the support images, it generated descriptive text that an LLM condensed into real semantic labels, effectively injecting meaning back into the pipeline.

Proposed Method



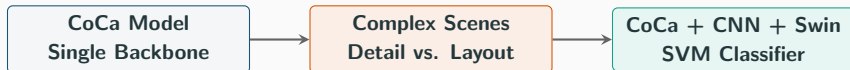
The Setup & The Flaw

Even with semantic labels, our approach was still purely Zero-Shot, leaving the explicitly provided 5-shot visual data completely untouched. Furthermore, standard CLIP is trained for global semantics and struggles to grasp dense, local image details.

The Solution

We fixed this by introducing a **Linear Probe** to actively train on the support features. Simultaneously, we replaced CLIP with **CoCa**. Since CoCa is forced to generate captions from scratch, it acts as a magnifying glass, extracting much denser and finer local details for a stronger visual foundation.

Proposed Method



The Setup & The Flaw

Relying on a single backbone (CoCa) created a new bottleneck. Our dataset consists of complex indoor and outdoor scenes. A single model inevitably struggles to balance the recognition of tiny local textures (like wall patterns) with the global spatial layout of the room.

The Solution

We evolved the pipeline into a **multi-model ensemble**. We concatenated CoCa's features with a **CNN** (for local patterns) and a **Swin Transformer** (for global structures). This massive, holistic vector was then classified using an **SVM**, finding the optimal margin without overfitting.

Results and Challenge Outcome

Results and Challenge Outcome

Ablation Study Results

Our systematic approach allowed us to measure the exact impact of each architectural upgrade on the 5-Way 5-Shot classification task:

- **Baseline (Standard CLIP Zero-Shot):** $\sim 20.0\%$ (*Equivalent to random guessing due to numeric labels*)
- **Semantic Rescue (CLIP + LLaVA Text Vectors):** $\sim 80.0\%$ (*Proves that restoring natural language context solves the primary bottleneck*)
- **Backbone Upgrade (CoCa + LLaVA + Linear Probe):** $\sim 90.0\%$ (*Generative details from CoCa drastically outperform standard CLIP*)

The Final Configuration

- **Visual Extractors:** CoCa + ConvNeXt + Swin Transformer.
- **Semantic Context:** LLaVA.
- **Classifier:** Support Vector Machine (SVM).

Peak Accuracy

94.60%

Scientific Validation

By concatenating generative features, local CNN patterns, and global Transformer structures, and classifying this high-dimensional space with an SVM, we achieved a near-perfect classification margin, completely mitigating low-data overfitting.

Results and Challenge Outcome

Phase	Configuration	Accuracy	Std Dev	Milestone / Impact
Step 1	Standard CLIP (Zero-Shot)	21.40%	$\pm 8.60\%$	Fails due to numeric labels.
Step 1b	CLIP + LLaVA (Text-Enriched)	82.80%	$\pm 10.81\%$	Semantic meaning restored.
Step 2	CLIP + LLaVA + Linear Probe	91.00%	$\pm 7.58\%$	Support set explicitly learned.
Step 2b	CoCa + CLIP Probe Tuning	93.80%	$\pm \text{N/A}$	CoCa extracts finer visual details.
Step 3	CoCa + ConvNeXt (CNN)	94.80%	$\pm 6.46\%$	Micro-textures captured.
FINAL	Ensemble (2560-dim) + Linear SVC	96.93%	$\pm 0.38\%$	Maximum margin, zero overfitting.

Why the Final Ensemble with Linear SVC?

The transition to the final **Linear SVC (C=0.50)** on a 2560-dimensional latent space didn't just maximize the absolute accuracy (**96.93%**); it completely stabilized the model. The standard deviation plummeted from $\sim 10\%$ to **0.38%**, proving the SVM perfectly handles high-dimensional concatenation without overfitting the 5-shot data.

Conclusion and Future Work

Conclusion & Future Work

- **Conclusion:** We successfully demonstrated that the limits of zero-shot classification on anonymized datasets can be systematically broken. By enriching query context via LLaVA and fusing diverse, complementary vision models (CoCa, CNN, Swin) through a linear SVM, we solved both the semantic gap and the few-shot overfitting dilemma.
- **Future Directions:**
 - **Dynamic Modality Weighting:** Implement attention mechanisms to automatically adjust each backbone's importance based on specific image complexity.
 - **Model Quantization:** Apply 4-bit (QLoRA) quantization to LLaVA to reduce inference VRAM footprint, making the pipeline highly deployable.
 - **Extreme Few-Shot Constraints:** Evaluate the ensemble's robustness on substantially harder scenarios (e.g., 20-way, 1-shot benchmarks).

Individual Contribution

Individual Contributions

Elia Mirafiori: Theory & Coordination

- **Foundations:** Established main theoretical guidelines.
- **Coordination:** Spearheaded the initial project phase.
- **Support:** Directed testing cycles & workload distribution.

Filippo Segalla: Compute & Dev

- **Hardware:** Provided compute power for heavy LLaVA tasks.
- **Coding:** Developed robust pipelines and the final SVM integration.
- **Cooperation:** Managed resource-intensive processes.

Vittorio Pettene: Evaluation & Benchmarks

- **Testing:** Executed main LLaVA validation loops.
- **Ablation:** Proved LLaVA outperforming basic zero-shot architectures.
- **Infrastructure:** Shared key high-performance environments.

Tommaso Delucchi: Vision & Dissemination

- **Vision Models:** Tested and evaluated CNN and Swin Transformers.
- **Integration:** Investigated model fusion techniques.
- **Materials:** Drafted technical slides & documentation.